

Argumentative Language in LLM: An Exploration

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Overview

Grid Search Space

Axis	Values
Models	Qwen2.5-7B, Llama3.1-8B, Mistral-7B, Gemma3-12B
Listener methods	direct_label, meta_label, CoT, self_rep_conf, log_probs
Speaker formats	full_sentence, three_blanks
Temperature	0.7 (fixed)
Runs per item	10
Total calls	6,000 (listener) + 3,200 (speaker) = 9,200

Two Experiments

	Listener	Speaker
Task	Infer who said it	Produce an utterance
Input	Exam data + sentence	Exam data + framing goal
Output	Student/Teacher/Examiner	OQ : IQ : ADJ
Conditions	high, low, info	high, low
Human N	242	121 (5×12)

Major Conclusions

- Best listener** Qwen2.5-7B × meta_label (JS=0.038)
Only combination balanced across all 3 conditions. Info=37% matches
- Best speaker** Gemma3-12B
Most human-like utterance diversity. Uses *most* for high, *some* for low.

Key Insights: Two Asymmetries

1. LLMs can do better production than comprehension. The production results are more human-like. Not too surprising given the nature of generative models. On the other hand, comprehension involves pragmatic reasoning over speaker's goal which might be harder for LLMs? Can this be generalized?
2. Among comprehension failures, LLMs are much better at detecting negative framing (low) than positive/promotional framing (high), which is the opposite of human pattern. This suggests a specific weakness in detecting positive framing strategies rather than a general failure to understand framing.

Experiment 1: Listener Side

Can LLMs infer which speaker type produced an utterance?

Core Prompt for the Listener Side (Shared by All Methods)

```
Background: A school recently administered an exam with {total_questions} questions to {n_students} students.
Exam results: {exam_data}

Three types of people described the results:
- Student: wants the exam to sound harder (as if students had a low chance of success), because this makes her own performance appear better by comparison.
- Teacher: wants the exam to sound easier (as if students had a high chance of success), because this reflects positively on teaching quality and students' preparedness.
- Examiner: aims to provide an objective and neutral description of the exam, without making it sound particularly hard or easy.

Someone said: "{sentence}"

[method-specific suffix appended here]
```

- Design principles:**
- Role descriptions are **exact copies** from experiment
 - Exam data formatted as "Student 1: 4/1"
 - Only the suffix varies by probing method

Model	Params
Qwen2.5	7B
Llama3.1	8B
Mistral	7B
Gemma3	12B

Grid: 5 methods × 4 models × 30 items × 10 runs, temp=0.7 fixed, 6,000 total calls

Probing Method Suffixes

1. direct_label

Who is most likely to have said this report? Answer with one word only:
Student, Teacher, or Examiner.

2. meta_label

Who is most likely to have said this report? Answer with one word only:
high, low, or info.

3. chain_of_thought

Who is most likely to have said this report? First give a one-sentence
analysis (max 20 words), then answer.
Format:
Analysis: [your reasoning]
Answer: [Student/Teacher/Examiner]

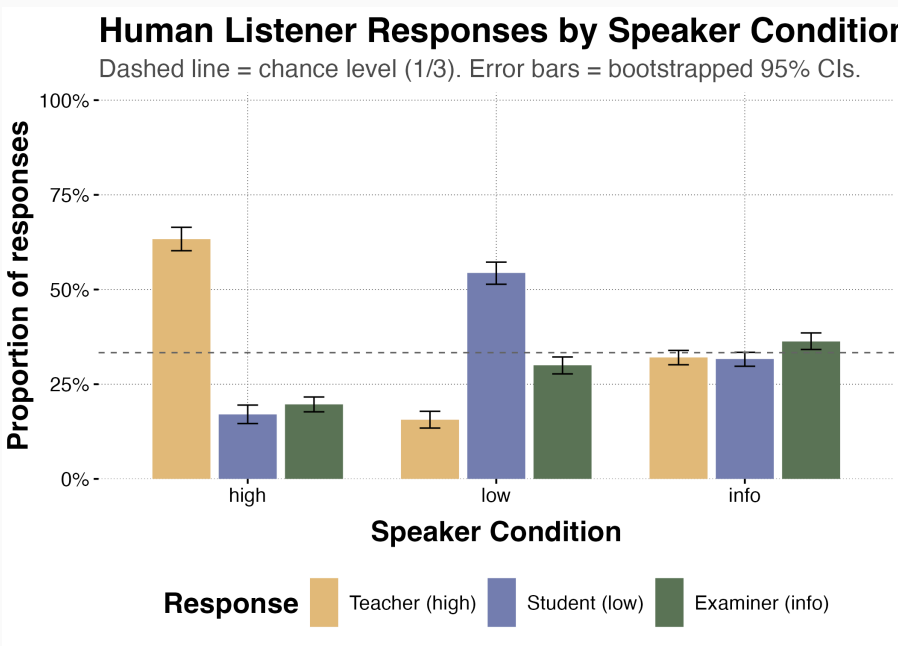
4. self_reported_confidence

Who is most likely to have said this report? Respond in this
format:
Answer: [Student/Teacher/Examiner]
Confidence: [0.0 to 1.0]

5. log_probs

Who is most likely to have said this report? Answer with one
Student, Teacher, or Examiner.
[+ logprobs=True, top_logprobs=5, max_tokens=5]

Human Baseline (N=242)

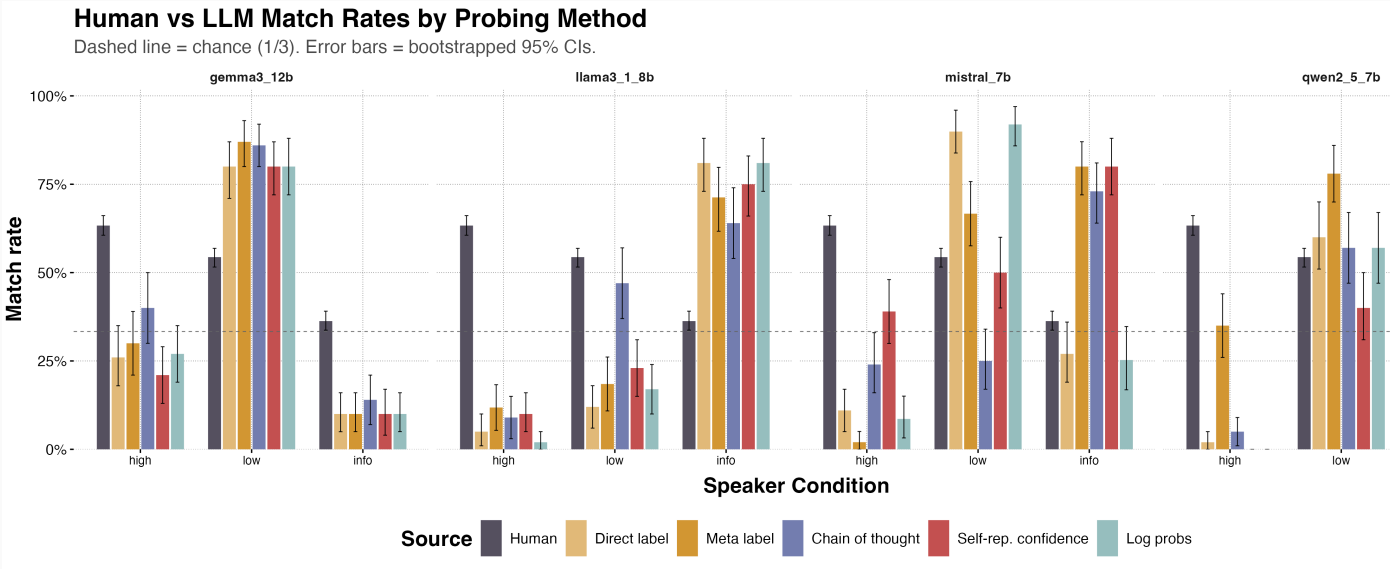


Condition	Match	95%
high	63.3%	[60.3
low	54.4%	[51.2
info	36.3%	[33.6

Core pattern:
high > low > info (chance)

Listeners detect argumentative framing, but
near chance.

Main Result: Human vs LLM Match Rates



Bars: Human (dark) + 5 probing methods. Error bars = bootstrapped 95% CIs. Dashed = chance (1/3).

Main Result: Key Findings

- high

is generally hard for LLMs
No combination reaches human 63%. Best: Gemma3 x CoT = 40%
- low

is over-detected
Gemma3 exceeds humans (80-87% vs 54%). LLMs are biased toward difficulty framing
- info

shows a model split
Llama/Qwen over-predict info (64-84%). Gemma under-predicts (10-14%)
- There is an interaction between model families and probing methods**
Some models (Gemma/Llama) do not show variation to different probing methods that change the data pattern qualitatively, some(Qwen/Mistral) do

Top 5 Combinations (JS Divergence)

#	Model	Method	JSD	H	L
1	Qwen2.5	meta_label	.038	35%	78%
2	Gemma3	CoT	.082	40%	86%
3	Mistral	self_rep	.112	39%	50%
4	Gemma3	log_probs	.120	27%	80%
5	Gemma3	direct	.122	26%	80%

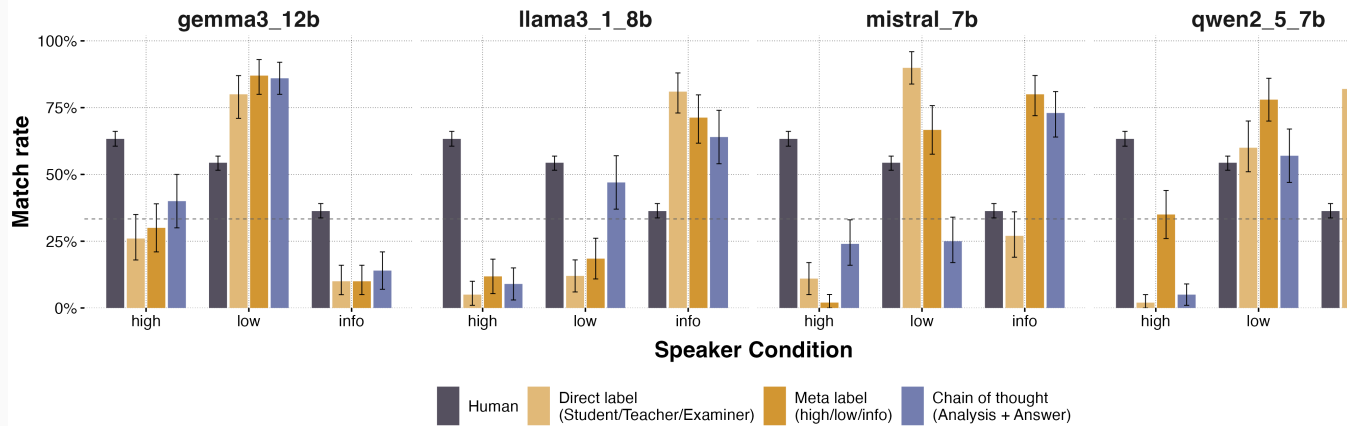
Winner

Qwen2.5 x meta_label
Only combination with balanced performance across all 3 conditions. It matches human 36%.

Focused: Direct Label vs Meta Label vs Chain of Thought

Direct Label vs Meta Label vs Chain of Thought

Comparing response format (role names vs abstract labels) and reasoning (CoT) effects.



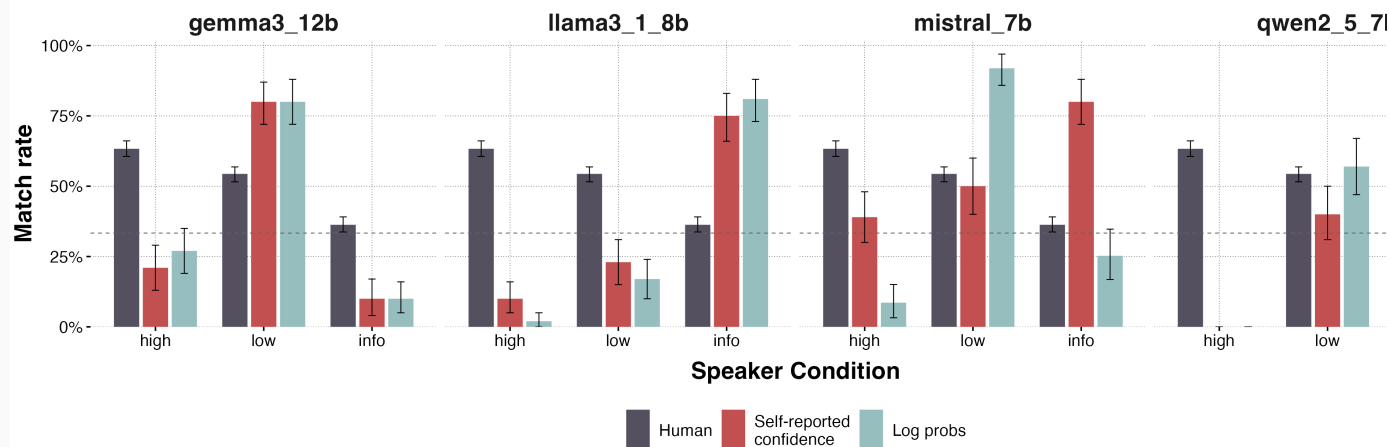
- **Qwen2.5:** Meta_label significantly better (JSD=0.038 vs 0.213). High: 2%→35%; info: 82%→37% (matches human 36%)
- **Gemma3:** CoT achieves best high-condition performance across all combinations (40%). Direct and meta similar
- **Llama3.1:** All methods struggle; CoT is best but high remains at 9%
- **Mistral:** CoT improves high (24%) but trades off with low (25% vs 90% for direct_label)

Effect of probing method is highly model-dependent. No single method dominates across all models.

Focused: Self-Reported Confidence vs Log Probs

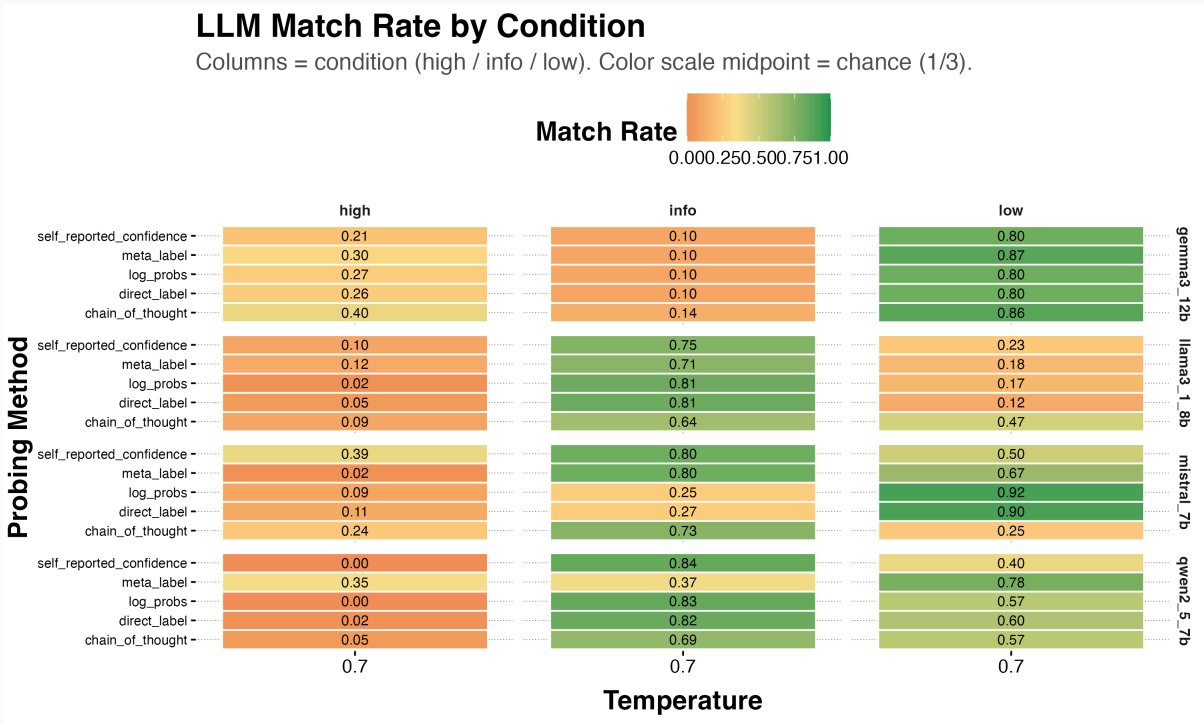
Self-Reported Confidence vs Log Probs

Both methods use the same prompt (direct_label format). Does the output format affect classification?



- Except for Mistral, self-reported confidence and log probs are highly correlated.
- But log_probs is tricky in term of API and model support. This means if we dont have that API support, we can just ask for confidence and get a similar signal. And maybe we should use natural language labels (e.g. very confident than numerical values for confidence)

Per-Condition Heatmap



Color midpoint = chance (1/3). Green = above chance. Red = below chance.

Open question: Why Does the High Condition Fail?

Human listeners identify Teacher (high) framing at 63% — the easiest condition. But LLMs max out at 40%.

- **Asymmetric framing detection:** LLMs detect "make it sound hard" (low) much better than "make it sound easy" (high)
- **Human is good at both and likely use the similar reasoning strategies,** but LLMs are overestimating low. The source of this asymmetry is an open question.

	Human	Best LLM
high	63%	40% (Gemma/CoT)
low	54%	87% (Gemma/meta)
info	36%	37% (Qwen/meta)

Open question: Is this specific to small (7-12B) models, or a general L limitation? Phase 2 will test larger models.

Experiment 2: Speaker Side

Can LLMs produce strategically framed utterances?

Speaker Side: Design & Prompt

- **Task:** Given exam results + framing goal, produce:
"*{OQ} of the students got {IQ} of the answers {ADJ}*"
- **2 framings:** high and low
- **20 arrays** (5 students × 12 questions)
- **10 runs** per item, temp=0.7
- **3,200 calls** total

Grid dimension: 2 output formats

full_sentence:
...Answer in the exact format only:
[OQ] of the students got [IQ] of the answers [ADJ]

three_blanks:
...Answer with exactly three words
separated by commas: [OQ], [IQ], [ADJ]

Core prompt (shared):

Background: A school recently administered an exam with 12 questions to 5 students.
Exam results: {exam_data}

Your task: Describe these exam results using a sentence of the exact format:
"[OQ] of the students got [IQ] of the answers [ADJ]"

where:

- OQ: none, some, most, or all
- IQ: none, some, most, or all
- ADJ: right or wrong

The sentence must be truthful.

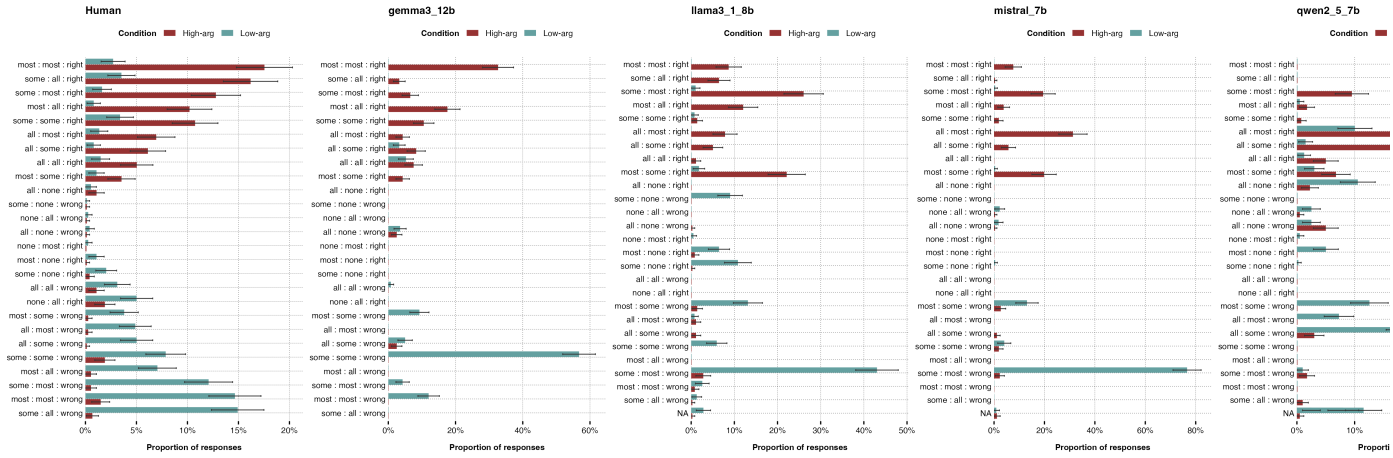
Describe these results so as to make it appear as if there is a {framing} rate without lying.

Invalid Rate	full_sentence	three_blanks
Gemma3	0.0% invalid	0.5%
Qwen2.5	0.8%	0.0%
Llama3.1	11.8%	2.0%
Mistral	47.5%	31.5%

Speaker: Human Baseline & LLM Comparison (Pooling from both output forma

Speaker Production: Utterance Choice Distribution (5 x 12)

Bars show high-arg (red) and low-arg (teal) framing. Ordered by human High-Low difference.



Utterances sorted by human high-low difference. Red = high-arg, teal = low-arg. Same ordering across all panels.

Speaker: Closeness to Human Data

Model	Format	High: % right (human: 93%)	Low: % wrong (human: 74%)	JSD high	JSD low	Mean JSD
Gemma3	full_sent	95%	93%	0.128	0.280	0.204
Llama3.1	full_sent	84%	75%	0.120	0.327	0.223
Llama3.1	3 blanks	99%	82%	0.215	0.348	0.282
Gemma3	3 blanks	95%	90%	0.152	0.451	0.302
Mistral	full_sent	95%	99%	0.227	0.438	0.332
Mistral	3 blanks	87%	98%	0.255	0.422	0.339
Qwen2.5	full_sent	87%	51%	0.315	0.432	0.374
Qwen2.5	3 blanks	91%	79%	0.316	0.477	0.397

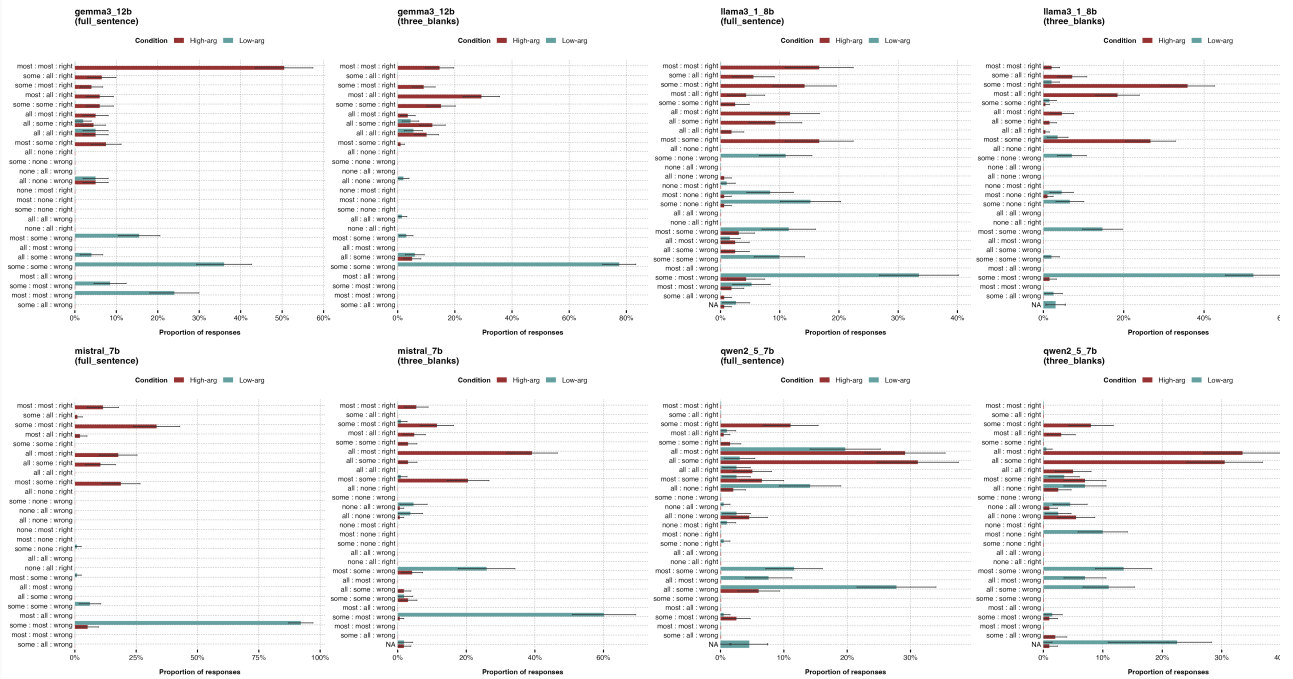
Best speaker Gemma3 (full_sentence)
Mean JSD = 0.204. Closest ADJ proportion to human utterance diversity to human.

- **ADJ choice:** All models get the d
Qwen weakest on low (51% wrong
full_sentence)
- **JSD low > JSD high:** Low framing
distributional divergence across a
- **Format effect:** Minimal for Gemm
Llama slightly better with full_sent

Speaker: Full Sentence vs Three Blanks

Output Format Comparison: Full Sentence vs Three Blanks

Does output format affect utterance choice distribution?



Same utterance ordering. Left column = full_sentence, right = three_blanks. Mistral has high invalid rate in full_sentence (47.5%) due to paraphrasing ("correct" instead of "right", dropped function words).

Speaker-Side Summary

The Puzzle

- **Speaker side:** All models can *produce* high-framing utterances (ADJ=right 89-95%)
- **LLMs do high framing speaker better than low-framing speaker**
Maybe this is because of the positive bias that give the feeling that LLMs overly agree and that why good at positive framing?
- **Listener side:** No model can *detect* high-framing utterances (max 40% vs human 63%)
- LLMs understand the strategy but can't infer intent from the output

Interpretation: The difficulty is in *pragmatic inference* about speaker goals, not in understanding the framing strategy itself. Production is easier than comprehension for LLMs — the reverse of many human language tasks.

	Speaker (produce)	Listener (detect)
high	ADJ=right 89-95%	Match 7-40%
low	ADJ=wrong 65-99%	Match 24-87%

Best speaker model: Gemma3 (most human-like OQ distribution)
Best listener model: Qwen2.5 x meta_label (JS=0.038)

Analysis Details

Listener Metrics

- **Match rate:** LLM label = ground-truth condition. Bootstrapped 95% CIs
- **Jensen-Shannon divergence:** LLM vs human response distribution. Lower = closer
- **5 probing methods:** direct_label, meta_label, CoT, self_rep_confidence, log_probs

Speaker Metrics

- **Utterance distribution:** OQ:IQ:ADJ proportions vs human
- **ADJ choice:** right/wrong proportion by framing
- **2 output formats:** full_sentence vs three_blanks

Infrastructure

- NVIDIA A6000 (48GB), Ollama v0.18.3
- Listener: 6,000 calls (~45 min)
- Speaker: 3,200 calls (~15 min)
- Temperature: 0.7 fixed
- 10 runs/item, uncertainty via bootstrap

Invalid Rates

Model	Listener	Speaker
Gemma3-12B	0%	0.2%
Qwen2.5-7B	0%	0.4%
Llama3.1-8B	1.4%	6.9%
Mistral-7B	1.1%	39.5%